

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

May 2028

College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Researching and implementing Java-based utilization optimizations for Amazon Redshift

Capital One

June 2026 – August 2026

Software Engineering Intern

Richmond, VA

- Rebuilt the legacy UI of a test-data workflow builder in React, Electron, and TypeScript, adding Jest-tested workflow validation and a React-Flow editor; rated **90%** satisfaction vs. the old tool
- Built a precomputed tree-style workflow search index to minimize DB queries under access limits, distilling popular workflows from a **270K+ row** PostgreSQL table via Python/SQLAlchemy and Redis
- Cut workflow generation time **65%** to under 15s with a Claude skill using sub-skill decomposition and Python search/validation tools, guided by CloudWatch + OpenSearch latency logs

Amazon (via UMD App Development Contracting)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Shipped an Amazon Leo space-research tool adopted into Leo's ecosystem, deployed on Docker + AWS Fargate via GitHub Actions, with AWS CDK (TypeScript) infra so new engineers can contribute
- Sped up a pipeline routing 6,000+ scraped research documents to inference by **34%** with Kinesis streams feeding Lambda processors, exposed to the app via API Gateway
- Automated fine-tuning of an AWS Bedrock source-relevancy model with EventBridge cron triggers and decision gates that skip redundant retraining; added X-Ray tracing and SNS failure alerts

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Developing automated training for PhysTwin, a physics-based digital twin, with a PhD team targeting ICLR
- Engineered a Python pipeline rendering AnyGrasp multi-camera grasp candidates as Open3D overlays for a Qwen3-VL-8B vision-language model, letting researchers isolate scene components
- Reduced research-environment setup time **30%** with uv dependency pinning and Docker, plus psutil time/memory profiling and Pytest suites to locate bottlenecks

TerpLabs

June 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded and scaled a **110+ member** campus product studio shipping 5 live products, run as agile teams
- Developed Prisma-based SQL querying with Redis query caching for Tortuga, a class scheduler that drew **20,000+ visits** in its first 30 days, to reduce DB load and latency
- Building a RAG meal-suggestion feature for the official UMD app with GPT-4o mini over ChromaDB + SentenceBERT recipe search; **93%** meal satisfaction among internal testers (in progress)

Projects

IEX Matching Engine Simulation | C++, STL, POSIX

July 2026 – August 2026

- Implemented a C++ order-book simulator on historical IEX data with a contract-based plug-in design, so alternative approaches can be swapped in with correctness enforced
- Designed paginated hash-based storage for live order-book data with an in-memory LFU cache for hot tickers, evaluated for drift against IEX historical book data
- Benchmarked a time-based adaptive LSM store against a B+ tree for simulation logs, measuring throughput and latency with rdtscp and hardware utilization with Linux perf

Technical Skills

Languages: Python, Java, C++, C, TypeScript/JavaScript, SQL

AWS: Lambda, Kinesis, Fargate/ECS, API Gateway, EventBridge, Bedrock, CDK, CloudWatch, X-Ray, SNS, IAM

Backend & Data: Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML & Tools: PyTorch, Hugging Face, RAG, Open3D, Docker, GitHub Actions, Linux, Pytest, Jest, uv, perf

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)